ZHENG KUANG

NHC Key Laboratory of Medical Immunology, Peking University, Beijing, China
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EDUCATION

Peking University Beijing, China
Master's degree in Immunology, School of Basic Medical Sciences Sept. 2023– Jun.2026
- GPA: 3.77/4.00
- Relevant Courses: Molecular Immunology, Molecular Biology Techniques, Omic big data analysis and application.

Naval Medical University Shanghai, China
Bachelor's degree in Traditional Chinese Medicine, School of Basic Medical Sciences Sept.2015– Jun. 2020
- Outstanding Graduate Award
- Relevant Courses: Biochemistry, Molecular Biology, and Medical Immunology.

RESEARCH PROJECTS

NHC Key Laboratory of Medical Immunology, Peking University Beijing, China
Master's Thesis Research, Supervisor: Prof Pingzhang Wang Sept. 2023– Jun.2026
• Investigating tumour-associated fibroblast subtypes in pancreatic cancer and immune–stroma interactions.
• Built a pan-cancer cross-study single-cell fibroblast annotation method.
• Performed flow cytometry and orthotopic tumour models for validation.
• Manuscript in preparation.

Shanghai Immune therapy Institute, Shanghai Jiaotong University Shanghai, China
Research Assistant Project, Supervisor: Prof Zhi Liu Jan. 2023 – Jul.2023
• Investigated the tissue-repair functions of regulatory T cells(Tregs) in skin tissue.
• Performed molecular and cellular assays, including flow cytometry, qPCR, and Western blotting, for immune cell phenotyping and functional validation.

Department of Bioinformatics, Shanghai Center for Translational Medicine Shanghai, China
Undergraduate research project, Supervisor: Prof Tengjiao Wang and Dr Mingzhen Yings Jun. 2017– Jun. 2020
• Discovered ICI response biomarkers in small cell lung cancer by analyzing genomic variants and bulk RNA-seq data.
• Performed differential expression profiling, pathway enrichment, and survival modeling.
• Contributed to two publications.

EXPERIMENTAL SKILLS

- **Molecular Cloning & Plasmid Construction:** Knockdown and overexpression constructs.
- **Virus Package & Purification:** Lentivirus, Retrovirus, and Adenovirus.
- **CRISPR:** Base editing, Prime editing and CRISPR screening.
- **Protein Purification:** ÄKTA-based purification; protein expression in E. coli and 293F cells.
- **Immunology-Related Techniques:** Flow cytometry analysis and sorting, Primary T-cell isolation and activation, and PBMC isolation.
- **Immunotherapy:** Phage display, scFv-based CAR-T design, and Fc-fusion protein design, expression, and functional evaluation.
- **Cell Culture:** Culture of various tumour cell lines and HEK293 cells.
- **Animal Experiments:** IV and SC injections; orthotopic pancreatic cancer model and subcutaneous tumor models.
- **Microscopy:** Immunofluorescence (IF), immunohistochemistry (IHC), and H&E staining.
- **Protein and RNA Experiments:** Western blotting and RNA extraction, reverse transcription, RT-qPCR.
- **Scientific Graphics:** FlowJo, ImageJ, Adobe Illustrator, Photoshop, and GraphPad Prism.

BIOINFORMATICS SKILLS PROFICIENT IN R, LINUX AND PYTHON

- **Programming and Computing**: R (Bioconductor, tidyverse, Seurat), Python (pandas, scikit-learn, Scanpy), Linux/Bash Shell, and High-Performance Computing (HPC) with Slurm.
- **Bulk Omics and Genomic Analysis**: Bulk RNA-seq, ATAC-seq, ChIP-seq, proteomics, whole-exome sequencing (WES), and CRISPR screen analysis.
- **Single-Cell & Spatial Omics**: scRNA-seq, single-cell immune repertoire analysis, and spatial transcriptomics.
- **Computational & Statistical Analysis**: Differential gene expression analysis, GO/GSEA, ssGSEA, survival analysis, Cox regression, WGCNA, pseudotime analysis, gene regulatory network analysis (SCENIC), motif

analysis, copy-number inference (inferCNV), cell–cell communication and spatial deconvolution (Cell2location), and CRISPR screen analysis (MAGeCK).

PUBLICATIONS

- Yuzhe H, Ling T, **Zheng Kuang**, ... & Pingzhang Wang. Mutual Exclusion Analysis Shows that DUSP9 Negatively Regulates PD-L1 Expression and Acts as a Target to Enhance Anti-PD-1 Efficacy. **Advanced Science**. (2026, third author)
- Huan L*, Audre A*, **Zheng Kuang**, ... & Chuan Chen. Endogenous Ceramide 24:1 Constrains Th17-Driven Neutrophilic Inflammation by Antagonizing EP2 Signalling. **Advanced Science**. (2026, second author)
- Wang L*, **Zheng Kuang***, Zhang D, ... & Tengjiao Wang. Reactive oxygen species in immune cells: A new antitumor target. **Biomedicine & Pharmacotherapy**. (2021, Co-first author)
- **Zheng Kuang**, Mingzhen Y. Current landscape of hyper-progression on immune checkpoint inhibitor therapy. **Chinese Clinical Oncology**. (2020, first author)

HONORS & AWARDS

- Outstanding Graduate Award, Peking University (2026)
- Award for Scientific Research, Peking University (2025)
- Outstanding Medical Student Award, Peking University (2025)
- Award for Academic Excellence, Peking University (2024)

REFEREES

- Prof Pingzhang Wang, NHC Key Laboratory of Medical Immunology, Peking University, wangpzh@bjmu.edu.cn
- Prof Zhi Liu, Shanghai Immune therapy Institute, Shanghai Jiaotong University, zhiliu@sjtu.edu.cn
- Prof Tengjiao Wang, Dept. Bioinformatics, Shanghai Center for Translational Medicine, tjwang@smmu.edu.cn
- Dr MingZhen Ying, Dept. Oncology, Shanghai ChangHai Hospital, madge_ying@126.com